3. Outer Measures
REAL VARIABLES I
Fall 2026

3.1 Outer Measures

Our goal is to assign a ”measure” to a ”most” subsets of a space X. Denoting 2^X the collection of subsets of a set X, we would like a to define a functional $\mu : 2^X \to \bar{R}_+$ that has the following properties:

- **non-negative:** $\mu(E) \geq 0$ for all $E \subset X$;
- **zero on empty set** $\mu(\emptyset) = 0$;
- **monotone** $\mu(A) \leq \mu(B)$ if $A \subset B$;
- **infinitely subadditive** $\mu(\bigcup_{i \in \mathcal{J}} A_i) \leq \sum_{i \in \mathcal{J}} \mu(A_i)$
- **infinitely additive on disjoint sets** $\mu(\bigcup_{i \in \mathcal{J}} A_i) = \sum_{i \in \mathcal{J}} \mu(A_i)$ if $A_i \cap A_j = \emptyset$ whenever $i \neq j$.

We will see, however, that, generally, these conditions are too strong: in many settings there are no useful functionals that satisfy all the properties above. As a first restriction, we only consider countably infinite subadditivity and additivity, (also known as σ-(sub)additivity).

Even then, these conditions are generally too restrictive, which motivates us to require fewer properties, thereby leading to the notion of an *outer measure.*

Definition 3.1.1 (Outer measure)**.** An *outer measure* on X is a function $\mu^* : 2^X \to [0, \infty]$ with the following properties:

1. $\mu^*(\emptyset) = 0$;
2. **σ-subadditivity:** for every sequence $(E_n \in 2^X)_{n \in \mathbb{N}}$ it holds $\mu^* \left(\bigcup_{n \in \mathbb{N}} E_n \right) \leq \sum_{n \in \mathbb{N}} \mu^*(E_n)$;
3. **monotonicity:** if $E \subseteq F \subseteq X$, then $\mu^*(E) \leq \mu^*(F)$.

The second property is called *countable subadditivity* or *σ-subadditivity.*

An outer measure is defined on all subsets of X, and the sets involved in the σ-subadditivity property need not be disjoint. For this reason we require only the inequality $\leq$ there. Furthermore, while important in its own right, follows automatically by σ subadditivity because and positivity. Indeed $F = E \cup (F \setminus E)$.

Example 3.1.2 (Counting outer measure)**.** For $E \subseteq X$, set

$$\mu^*(E) \coloneqq \begin{cases} \#E & \text{if } E \text{ is finite,} \\ \infty & \text{otherwise.} \end{cases}$$

Let us show that μ^* is an outer measure.

A slightly more involved example is a "weighted" version of the above construction.

Example 3.1.3 (Weighted counting outer measure)**.** Let $w\colon X \to [0, \infty]$. For $E \subseteq X$, define

$$\mu_w^*(E) \coloneqq \sum_{x \in E} w(x) \coloneqq \sup\left\{ \sum_{x \in F} w(x) : F \subseteq E, F \text{ finite} \right\}. \tag{1}$$

Then μ_w^* is an outer measure and is countably additive on disjoint sequences. Counting outer measure is the case $w(x) = 1$ for every $x \in X$.

Example 3.1.4 (Lebesgue outer measure)**.** For any $E \subset \mathbb{R}$ define

$$\mathcal{L}^*(E) \coloneqq \inf\Big\{ \sum_{j=0}^{\infty} b_j - a_j \ : \ \bigcup_{j=0}^{\infty} (a_j, b_j) \supseteq E \Big\}.$$

Next, we will see that the formula used to define $\mathcal{L}^*$ is a common way to define outer measures. The fact that $\mathcal{L}^*$ is indeed an outer measure will follow from general properties of this construction.

While the first two examples we saw are actually measures: they satisfy countable additivity on disjoint families, we will conclude this section by showing that the Lebesgue outer measure is **not** σ-additive on disjoint sets.

3.1.1 Generated Outer measures

Next, we illustrate a very common way to explicitly define outer measures.

Let $\mathcal{A} \subseteq 2^X$ contain $\emptyset$, and let $\ell\colon \mathcal{A} \to [0, \infty]$ satisfy $\ell(\emptyset) = 0$. We regard $\ell(A)$ as the "cost" of using $A \in \mathcal{A}$ in a cover of a set E.

Definition 3.1.5 (Outer measure generated by a cost)**.** For $E \subseteq X$, define the *outer measure generated by* ℓ as

$$\ell^*(E) \coloneqq \inf \left\{ \sum_{n \in \mathbb{N}} \ell(A_n) : (A_n \in \mathcal{A})_{n \in \mathbb{N}}, \quad E \subseteq \bigcup_{n \in \mathbb{N}} A_n \right\}. \tag{2}$$

If no countable family in $\mathcal{A}$ covers E, the infimum is ∞.

Let us show that the functional ℓ^* in (2) is an outer measure on X.

Proof. The constant sequence $A_n = \emptyset$ covers $\emptyset$ and has cost zero. Hence $\ell^*(\emptyset) = 0$.

If $E \subseteq F$, then every countable cover of F is a countable cover of E. Taking infima gives $\ell^*(E) \leq \ell^*(F)$.

It remains to prove countable subadditivity. Let $(E_n \in 2^X)_{n \in \mathbb{N}}$. If $\ell^*(E_n) = \infty$ for some n, the required inequality is immediate. Assume that $\ell^*(E_n) < \infty$ for every n, and fix $\varepsilon > 0$. For each $n \in \mathbb{N}$, choose a sequence $(A_{n,k} \in \mathcal{A})_{k \in \mathbb{N}}$ such that

$$E_n \subseteq \bigcup_{k \in \mathbb{N}} A_{n,k} \quad \text{and} \quad \sum_{k \in \mathbb{N}} \ell(A_{n,k}) \leq \ell^*(E_n) + \varepsilon 2^{-n-1}.$$

The family $(A_{n,k})_{(n,k) \in \mathbb{N}^2}$ is countable and covers $\bigcup_{n \in \mathbb{N}} E_n$. Therefore

$$\ell^* \left(\bigcup_{n \in \mathbb{N}} E_n \right) \leq \sum_{n \in \mathbb{N}} \sum_{k \in \mathbb{N}} \ell(A_{n,k}) \leq \sum_{n \in \mathbb{N}} \ell^*(E_n) + \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, this proves countable subadditivity. $\square$

The construction has the following extremal property: ℓ^* is the *largest* outer measure that is smaller than ℓ on $\mathcal{A}$.

Proposition 3.1.6 (Largest outer measure below a cost)**.** For every $A \in \mathcal{A}$, $\ell^*(A) \leq \ell(A)$.

Moreover, if ν^* is an outer measure on X such that $\nu^*(A) \leq \ell(A)$ for every $A \in \mathcal{A}$, then $\nu^*(E) \leq \ell^*(E)$ for every $E \subseteq X$.

Proof. To show that $\ell^*(A) \leq \ell(A)$ note that the set A together with countably many copies of $\emptyset$ is a countable cover of A.

Next, suppose $\nu^*(A) \leq \ell(A)$ for every $A \in \mathcal{A}$. Fix $E \subseteq X$ and let us show that $\nu^*(E) \leq \ell^*(E)$. For any countable cover $(A_n \in \mathcal{A})_{n \in \mathbb{N}}$ cover E we have, by monotonicity and countable subadditivity,

$$\nu^*(E) \leq \nu^* \left(\bigcup_{n \in \mathbb{N}} A_n \right) \leq \sum_{n \in \mathbb{N}} \nu^*(A_n) \leq \sum_{n \in \mathbb{N}} \ell(A_n).$$

Taking the infimum over all such covers proves Proposition 3.1.6. $\square$

Example 3.1.7 (A generated outer measure larger than weighted counting)**.** Enumerate $\mathbb{Q}$ as $(q_n)_{n\geq 1}$, and define $\ell\colon 2^{\mathbb{R}} \to [0,\infty]$ by

$$\ell(E) = \begin{cases} \displaystyle\sum_{q_n \in E} 2^{-n}, & E \text{ is countable,} \\ \infty, & E \text{ is uncountable.} \end{cases}$$

Also define the weighted counting outer measure

$$\nu^*(E) = \sum_{q_n \in E} 2^{-n}.$$

Then $\nu^*(E) \leq \ell(E)$ for every $E \subseteq \mathbb{R}$, so Proposition 3.1.6 gives $\nu^* \leq \ell^*$.

If E is countable, the one-set cover of E gives $\ell^*(E) \leq \ell(E) = \nu^*(E)$. Hence $\ell^*(E) = \nu^*(E)$. If E is uncountable, every countable cover of E contains an uncountable set. Every such cover has infinite cost, and therefore $\ell^*(E) = \infty$. Consequently,

$$\ell^*(E) = \begin{cases} \displaystyle\sum_{q_n \in E} 2^{-n}, & E \text{ is countable,} \\ \infty, & E \text{ is uncountable.} \end{cases}$$

In particular, ℓ^* is strictly larger than ν^* on every uncountable subset of $\mathbb{R}$.

In general, according to (2), it can happen that $\ell^*(A) < \ell(A)$, that is there is a way to cover a set $A \in \mathcal{A}$ in a more optimal way than described by the cost functional ℓ. Mostly, however, we are interested in using ℓ^* as a way to *extend* ℓ. The following condition precisely characterizes when the ℓ^* coincides with ℓ on the original collection $\mathcal{A}$.

Proposition 3.1.8 (Extension of a cost)**.** The identity

$$\ell^*(A) = \ell(A), \qquad A \in \mathcal{A},$$

holds if and only if the following covering inequality holds: for every $A \in \mathcal{A}$ and every sequence $(A_n \in \mathcal{A})_{n\in\mathbb{N}}$,

$$A \subseteq \bigcup_{n\in\mathbb{N}} A_n \quad \Longrightarrow \quad \ell(A) \leq \sum_{n\in\mathbb{N}} \ell(A_n). \tag{3}$$

When $\mathcal{A} = 2^X$, condition (3) is equivalent to monotonicity and countable subadditivity of ℓ.

Proof. The proof is omitted. $\square$

3.1.2 Lebesgue outer measure in dimensions one and two

In this section we focus on the Lebesgue outer measure on $\mathbb{R}^d$. For simplicity, we will focus on the case $d = 1$ and $d = 2$. Higher dimensional cases are notationally more complex but present no additional difficulties. An open rectangle in $\mathbb{R}^d$ is a set of the form

$$\prod_{j=1}^{d}(a_j, b_j), \qquad -\infty < a_j < b_j < \infty,$$

and its volume is $\mathrm{Vol}_d\left(\prod_{j=1}^{d}(a_j, b_j)\right) = \prod_{j=1}^{d}(b_j - a_j)$.

Definition 3.1.9 (Lebesgue outer measure)**.** For $E \subseteq \mathbb{R}^d$, define

$$\mathcal{L}_d^*(E) \coloneqq \inf\left\{\sum_{n\in\mathbb{N}} \mathrm{Vol}_d(R_n) : (R_n)_{n\in\mathbb{N}} \text{ are open rectangles and } E \subseteq \bigcup_{n\in\mathbb{N}} R_n\right\}. \tag{4}$$

The definition uses the construction in Definition 3.1.5. It is not yet automatic that a rectangle has outer measure equal to its usual length or area.

Proposition 3.1.10 (Lebesgue outer measure)**.** For $d \in \{1, 2\}$, the functional $\mathcal{L}_d^*$ is an outer measure on $\mathbb{R}^d$, and $\mathcal{L}_d(R) = \mathrm{Vol}_d(R)$ for all open rectangles.

The fact that $\mathcal{L}_d^*$ is an outer measure follows from Proposition 3.1.6, applied to the family consisting of the open rectangles and $\emptyset$, with cost Vol_d.

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